

Cross-chain γ

$\|b\|^2 \leq S$ is a kernel comparison; $\|b\|^2 < B_w$ waits on $q_N < 1$

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Abstract

Round 424 identified b as the μ -Parseval coefficient of the signed border σ . Both $\|b\|^2$ and S are the N -Christoffel energies of the same σ , in the μ -ONB and the $\mu - \nu$ -ONB of $P_{<N}$. On that space $\langle p, p \rangle_\mu \geq \langle p, p \rangle_{\mu-\nu}$ as soon as ν is a positive measure and the signed form is definite (tilde chain ok). Reproducing kernels then satisfy $K^{\mu-\nu} \succeq K^\mu$, hence $\|b\|^2 \leq S$. $B_w = S_{N-2} + \frac{5}{7}$ is S exactly when $q_N = 1$. The posted $\|b\|^2 < B_w$ is this SATZ plus $q_N < 1$. No RH claim.

Status. Lemma ?? is **reduced** (cofinal $q_N < 1$ remains). Proposition ?? is a finite theorem. On the 42-rung $q_N < 1$ census it *composes* to $\gamma < 1$. Ausgang B2_LE_S_SATZ / QN_BRIDGES_BW / DEAD_NEEDS_SLACK / COFINAL_OPEN. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The lemma

Lemma 1 (cross-chain γ). $\|b\|^2 < B_w = S_{N-2} + \frac{5}{7}$ as a cross-chain theorem, so $\gamma < 1$ and $\text{den} = 1 + \gamma - v_t \cdot s < 2$ even at $v_t \cdot s = 0$.

Verdict: REDUZIERT as posted. The sharp theorem is $\|b\|^2 \leq S$ (Proposition ??). $S < B_w$ iff $q_N < 1$. Cofinal $q_N < 1$ is the remaining gap (already the $v960$ census, not cofinal).

2 The common majorant

Let $P_{<N}$ carry the two inner products

$$\langle p, p \rangle_\mu = \int p^2 d\mu, \quad \langle p, p \rangle_{\mu-\nu} = \int p^2 d\mu - \int p^2 d\nu.$$

Whenever the tilde chain exists ($h_k > 0$ through the window), the second form is positive definite on $P_{<N}$. Since ν is a positive measure, $\langle p, p \rangle_\mu \geq \langle p, p \rangle_{\mu-\nu}$. The reproducing kernels of a finite-dimensional space then satisfy $K_N^{\mu-\nu} \succeq K_N^\mu$ (the unit ball of the smaller form is larger, so the Christoffel kernel is larger).

Proposition 1 (kernel Loewner). *If the tilde chain is positive through N , then for every finite atomic signed border $\sigma = \sum_a w_a \delta_{t_a}$,*

$$\|b\|^2 = w^\top K_N^\mu w \leq w^\top K_N^{\mu-\nu} w = S.$$

Over $\mathbb{Q}$: $\mu = (2, 1)$ at $(0, 1)$, $\nu = (1)$ at 1 , constants, $K^\mu = \frac{1}{3} \leq K^{\mu-\nu} = \frac{1}{2}$, Dirac at 0 has ratio $\frac{2}{3}$. At $k_z = 9$, the Rayleigh is 0.7264; the operator norm of the change-of-basis on $\text{col}(\Psi)$ is 1.0000.

The naive direction “ $L^2(\mu) \geq L^2(\mu - \nu) \Rightarrow \mu\text{-Parseval} \geq \text{tilde-Parseval}$ ” is the *wrong* way for measures: a larger form has a *smaller* kernel. That is why $\|b\|^2 \leq S$, not the reverse.

The full Parseval sum over all μ -degrees is not a tame σ -norm: σ sits on border atoms, typically off the μ -support, so $K_M^\mu(t, t)$ explodes as M approaches the discrete μ -degree.

3 Weighted Cauchy–Schwarz / Gram

Termwise $w_k = b_k^2/\rho_k$ remains unbounded (r424). The exact cross-identity is not $b_k = c F_k$ per index; it is the change-of-basis $b = Gf$ on $P_{<N}$, with $f_k = F_k/\sqrt{\eta_k}$ and $\|f\|^2 = S$. $\|G\|_{\text{op}} = 1$ on $\text{col}(\Psi)$ — a construction object (the identity map between the two norms). The true window border is *not* the maximizer: random borders have Rayleigh ~ 0.02 ; the one-atom maximizer-near direction at $k_z = 9$ has 0.75; $k = 5$ is the tight living ray 0.799. A uniform $\|G\|_{\text{op}} \leq 0.9$ is false ($\|G\|_{\text{op}} = 1$). The 0.80 census is the Rayleigh of *this* σ , not an operator bound.

4 Sharp form, reach, Lean

The sharp theorem is $\|b\|^2 \leq S$. On any window with $q_N < 1$, $S < B_w$ and therefore $\gamma < 1$. The 42-rung $q_N < 1$ census ($q_N \in [0.0015, 0.9805]$) plus Proposition ?? gives $\gamma < 1$ and, with the r423 formula, $\text{den} < 2$ as a composed finite statement on those rungs. Σ -boundedness and the reserve floor stay census/evidence; the cofinal edge is still conditional on cofinal $q_N < 1$.

Lean sketch (not this round): a named **Prop** on a finite-dimensional real inner-product space, **inner mu** $\geq$ **inner mutilde** (certificate: two Gram matrices, Cholesky of the difference $= \sum_j v_j p(y_j)^2$), then the kernel Loewner and the pairing inequality. $\mathbb{Q}$ -toy as an instance.

5 Kills

world / mutant	$\ b\ ^2 \leq S$	$q_N < 1$	$\gamma < 1$
core-42 + EXT + selected	yes	yes	SATZ (composed)
$k = 5$ (tight ray 0.799)	yes	0.023	kernel slack
dead χ	yes (0.62)	no (1.21, 1.33)	slack, not q_N
random border	yes (~ 0.02)	(chain ok)	far from max
atom reorder	yes	yes	measure, not order
scramble	n/a	n/a	tilde form not definite
unnorm b	n/a	—	$\gamma = 2.51 > 1$

Dead χ is the named witness that $\|b\|^2 \leq S$ does *not* by itself give $\|b\|^2 \leq B_w$ when $q_N > 1$.

6 What is a theorem

Proposition ?? is a theorem. Lemma ?? is not cofinal: it is this SATZ plus $q_N < 1$. On the named 42-rung census the composition is a theorem and $\text{den} < 2$ follows. No RH claim.